Type MS · Draw a line or curve of best fit through plotted points

Answer 1 of 6 · Question 2 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · Draw a line or curve of best fit through plotted points

Answer 1 of 6 · Question 2 part a.ii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 2 of 6 · Question 3 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

3	(a)	(i)	A (cornea and lens); B is incorrect as the iris does not play a role in focusing light C is incorrect as the iris does not play a role in focusing light D is incorrect as the pupil does not play a role in focusing light		1
	(b)	(i)	Points as follows: 150:2.7 200:3.2 250:2.3;		1
Clip (b)(i) with (b)(ii) and (b)(iii)		(ii)	Line of best fit missing out point at 150:2.7	Ignore lines that extended beyond points	1
		(iii)	150 lux/150:2.7/2.7 diameter;		1
			does not follow the pattern of other points/trend/not on the line of best fit;		1
		(iv)	As light intensity increases diameter decreases;	ORA	1
	(c)		Idea that distant/far away objects are blurred/difficult to see/ORA; as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much;	Reject can't see	1 1

Total question 3 = 8 marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 2 of 6 · Question 3 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 3 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a)	(i)	• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
(b)	(i)	axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 3 of 6 · Question 2 part b.ii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 4 of 6 · Question 5 part b.ii · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number					
5	(a)	(i)	$C_6H_{12}O_6$; $6H_2O$;		2
		(ii)	ATP breaks down/hydrolysed; to ADP and Pi;	Allow phosphate/P	2
	(b)	(i)	• suitable axis labels with units; • time on x axis; • suitable scale; • correct plots;		4
		(ii)	• suitable curve;		1
		(iii)	Five from; • mass of haemoglobin increases (with time/training at altitude); • increased number of red blood cells/more haemoglobin in the blood; • haemoglobin binds oxygen/red blood cells carry oxygen; • more oxygen transported around the body/in the blood; • to muscle (cells/tissue); • for aerobic respiration; • to release energy;		5

Total 14 marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 4 of 6 · Question 5 part b.ii · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 5 of 6 · Question 4 part a.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

Question number				
4	(a) (i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
	(b)	• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Draw a line or curve of best fit through plotted points

Answer 6 of 6 · Question 4 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	 Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%; • all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	MP1- accept Vmax not enough/no more substrate/protein to bind to active sites= 2 marks	2
					Total = 15

Type MS · Draw a line or curve of best fit through plotted points

Answer 6 of 6 · Question 4 part a.ii · 1 mark

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks